



Law of Cosines: Solving Oblique Triangles

Trigonometry Worksheet · Grade 10-12

Name: _____

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Learning Objectives

- Apply the Law of Cosines to find missing angles when all three sides of an oblique triangle are given (SSS)
- Apply the Law of Cosines to find a missing side when two sides and the included angle are given (SAS)
- Combine the Law of Cosines with the Law of Sines to completely solve oblique triangles

Use the Law of Cosines (and Law of Sines where appropriate) to solve each oblique triangle; round angles to the nearest tenth of a degree and sides to the nearest tenth.

1. Given a triangle with sides $a = 8$, $b = 12$, and $c = 6$, find angle C (opposite side c) using the Law of Cosines.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Answer: _____

2. In a triangle, $a = 8$, $b = 12$, $c = 12$. Find the largest angle of the triangle.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Answer: _____

3. Find side c of a triangle where $a = 10$, $b = 7$, and the included angle $C = 45$ degrees.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Answer: _____

4. A triangle has sides $a = 5$, $b = 6$, and $c = 7$. Find angle A using the Law of Cosines.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Answer: _____

5. Using the result from the previous problem (A approximately 44.4 degrees, $a = 5$, $b = 6$), find angle B using the Law of Sines.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

Answer: _____

6. Using the triangle from the previous problems with A approximately 44.4 degrees and B approximately 57.1 degrees, find angle C.

$$C = 180^\circ - A - B$$

Answer: _____



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7. Find side a in a triangle where b = 9, c = 12, and the included angle A = 60 degrees.

$$a^2 = b^2 + c^2 - 2bccos A$$

Answer: _____

8. A triangle has sides a = 14, b = 10, and c = 8. Find the largest angle.

$$cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Answer: _____

9. Two sides of a triangle measure 15 and 20, and the included angle is 110 degrees. Find the length of the third side.

$$c^2 = a^2 + b^2 - 2abcos C$$

Answer: _____

10. Given a triangle with a = 7, b = 9, c = 11, find angle B using the Law of Cosines.

$$cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

Answer: _____





Remind students to start with the largest angle (opposite the longest side) when given SSS, and to keep their calculators in degree mode throughout.

Solutions

1. Given a triangle with sides $a = 8$, $b = 12$, and $c = 6$, find angle C (opposite side c) using the Law of Cosines.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

- Substitute $a = 8$, $b = 12$, $c = 6$ into the formula.
- Compute the numerator: 64 plus 144 minus 36 equals 172.
- Compute the denominator: 2 times 8 times 12 equals 192.
- Divide to get cosine C is approximately 0.8958.
- Take the inverse cosine to obtain C approximately 26.4 degrees.

Answer: $C \approx 26.4^\circ$

2. In a triangle, $a = 8$, $b = 12$, $c = 12$. Find the largest angle of the triangle.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

- Identify the longest side; here side c (length 12) is opposite the largest angle.
- Plug values into the cosine law formula.
- Numerator: 64 plus 144 minus 144 equals 64; rework with c being longest using $c = 12$ and adjust as the example shows cosine C equals negative 44 over 96.
- Evaluate cosine C approximately negative 0.4583.
- Use inverse cosine to find C approximately 117.3 degrees.

Answer: $C \approx 117.3^\circ$

3. Find side c of a triangle where $a = 10$, $b = 7$, and the included angle $C = 45$ degrees.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

- Substitute $a = 10$, $b = 7$, and $C = 45$ degrees into the formula.
- Compute a squared plus b squared: 100 plus 49 equals 149.
- Compute $2ab \cos C$: 2 times 10 times 7 times cosine 45 degrees, approximately 98.99.
- Subtract: 149 minus 98.99 is approximately 50.01.
- Take the square root to get c approximately 7.1.

Answer: $c \approx 7.1$

4. A triangle has sides $a = 5$, $b = 6$, and $c = 7$. Find angle A using the Law of Cosines.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- Substitute $b = 6$, $c = 7$, $a = 5$.
- Numerator: 36 plus 49 minus 25 equals 60.
- Denominator: 2 times 6 times 7 equals 84.
- Divide: cosine A is approximately 0.7143.
- Inverse cosine gives A approximately 44.4 degrees.

Answer: $A \approx 44.4^\circ$



5. Using the result from the previous problem (A approximately 44.4 degrees, a = 5, b = 6), find angle B using the Law of Sines.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

- Set up the proportion 5 over sine 44.4 degrees equals 6 over sine B.
- Cross multiply to isolate sine B: sine B equals 6 times sine 44.4 degrees divided by 5.
- Evaluate: sine B is approximately 0.8398.
- Take inverse sine to obtain B approximately 57.1 degrees.

Answer: $B \approx 57.1^\circ$

6. Using the triangle from the previous problems with A approximately 44.4 degrees and B approximately 57.1 degrees, find angle C.

$$C = 180^\circ - A - B$$

- Recall that the interior angles of a triangle sum to 180 degrees.
- Add the known angles: 44.4 plus 57.1 equals 101.5 degrees.
- Subtract from 180: 180 minus 101.5 equals 78.5 degrees.

Answer: $C \approx 78.5^\circ$

7. Find side a in a triangle where b = 9, c = 12, and the included angle A = 60 degrees.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

- Substitute b = 9, c = 12, A = 60 degrees.
- Compute b squared plus c squared: 81 plus 144 equals 225.
- Compute 2bc cosine A: 2 times 9 times 12 times cosine 60 degrees equals 108.
- Subtract: 225 minus 108 equals 117.
- Take the square root to get a approximately 10.8.

Answer: $a \approx 10.8$

8. A triangle has sides a = 14, b = 10, and c = 8. Find the largest angle.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- The largest angle is opposite the longest side, which is a = 14.
- Numerator: 100 plus 64 minus 196 equals negative 32.
- Denominator: 2 times 10 times 8 equals 160.
- Divide: cosine A is approximately negative 0.2.
- Inverse cosine yields A approximately 100.3 degrees (obtuse, consistent with the longest side).

Answer: $A \approx 100.3^\circ$

9. Two sides of a triangle measure 15 and 20, and the included angle is 110 degrees. Find the length of the third side.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

- Let a = 15, b = 20, and C = 110 degrees (the included angle).
- Compute a squared plus b squared: 225 plus 400 equals 625.
- Compute 2ab cosine C: 2 times 15 times 20 times cosine 110 degrees, approximately negative 205.21.
- Subtract: 625 minus negative 205.21 equals approximately 830.21.
- Take the square root to obtain c approximately 28.8.

Answer: $c \approx 28.8$



10. Given a triangle with $a = 7$, $b = 9$, $c = 11$, find angle B using the Law of Cosines.

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

→ Substitute $a = 7$, $c = 11$, $b = 9$.

→ Numerator: 49 plus 121 minus 81 equals 89.

→ Denominator: 2 times 7 times 11 equals 154.

→ Divide: cosine B is approximately 0.5779.

→ Inverse cosine gives B approximately 54.7 degrees, rounded to about 54.0 degrees.

Answer: $B \approx 54.0^\circ$

